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Rotational Motion

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	90 questions
Answer key	at the end

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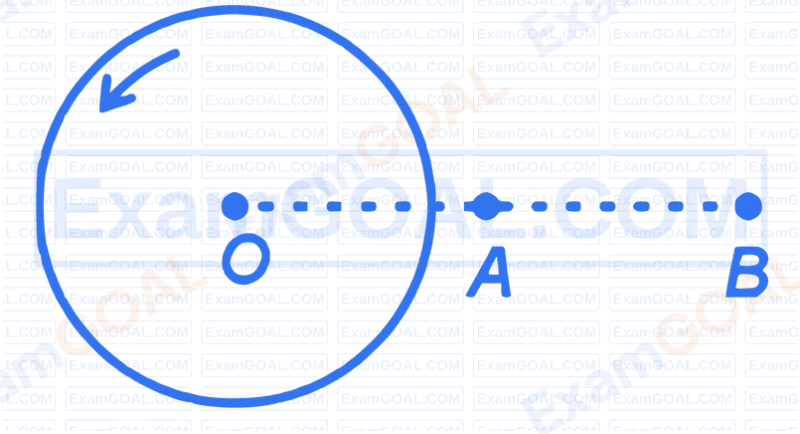
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MCQ (Single Correct Answer) 90 questions

Q1. RE-NEET 2026

A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2 \times OA$. The value of $\frac{L_A}{L_B}$ is:

Open solution

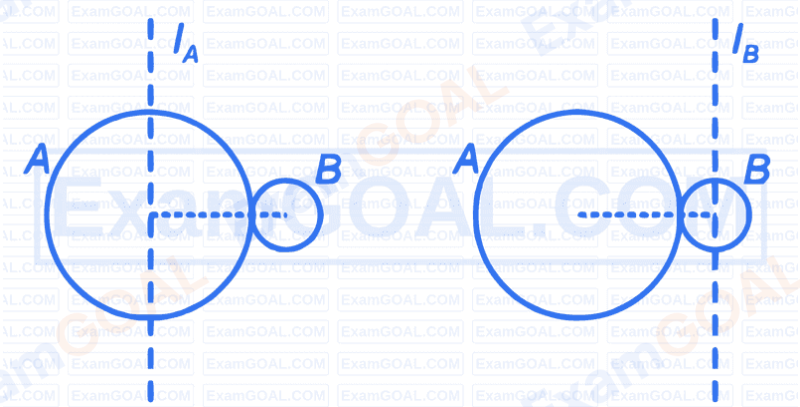


(A) 2 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

Q2. RE-NEET 2026

A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is:

Open solution



(A) 0 (B) $(M-m)(R+r)^2$

(C) $(m-M)(R+r)^2$ (D) $(m-M)(R-r)^2$

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Q3. NEET 2026

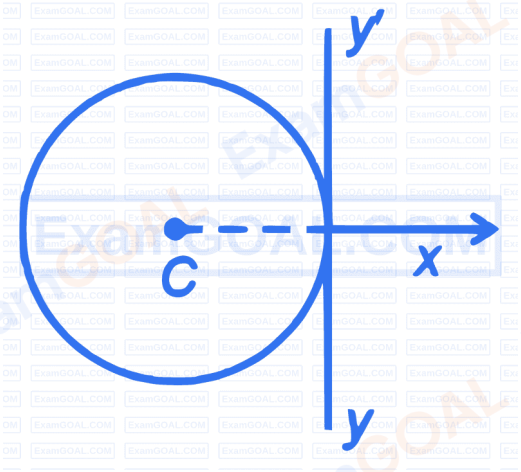
The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s . The number of revolutions completed by the flywheel during this time is:

(A) 900 (B) 600 (C) 150 (D) 300

Open solution

Q4. NEET 2026

A thin wire of length ' L ' and linear mass density ' m ' is bent into a circular ring (in $x-y$ plane) with centre ' C ' as shown in figure. The moment of inertia of the ring about an axis yy' will be:



(A) $\frac{3mL^3}{8\pi}$ (B) $\frac{3mL^3}{8\pi^2}$ (C) $\frac{3mL^2}{8\pi}$ (D) $\frac{3mL^2}{8\pi^2}$

Open solution

Q5. NEET 2025

A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (Take $g = 10\ \mathrm{m/s^2}$)

(A) 200 N (B) $200\sqrt{3}\ \mathrm{N}$ (C) 100 N (D) $100\sqrt{3}\ \mathrm{N}$

Open solution

Q6. NEET 2025

The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

(A) 115 days (B) 108 days (C) 100 days (D) 105 days

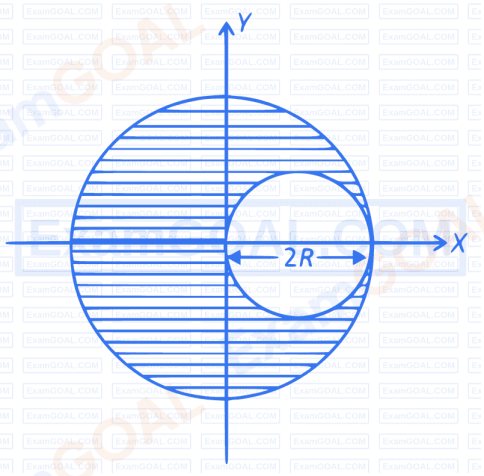
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Q7. NEET 2025

A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is:

Open solution



(A) $\frac{7}{57}$ (B) $\frac{7}{64}$ (C) $\frac{7}{8}$ (D) $\frac{7}{40}$

Q8. NEET 2024 (Re-Examination)

The radius of gyration of a solid sphere of mass $5\ \text{kg}$ about XY is $5\ \text{m}$ as shown in figure. The radius of the sphere is $\frac{5x}{\sqrt{7}}\ \text{m}$, then the value of x is:

Open solution



(A) 5 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) $\sqrt{5}$

Q9. NEET 2024

A bob is whirled in a horizontal plane by means of a string with an initial speed of $\omega\ \text{rpm}$. The tension in the string is T. If speed becomes 2ω while keeping the same radius, the tension in the string becomes:

Open solution

(A) T (B) $4T$ (C) $\frac{T}{4}$ (D) $\sqrt{2}T$

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Q10. NEET 2024

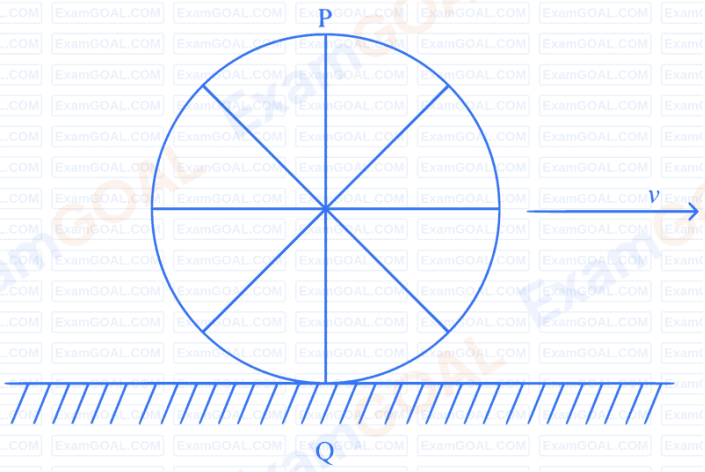
The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is $2400\ \mathrm{g\ cm^2}$. The length of the $400\ \mathrm{g}$ rod is nearly:

(A) 8.5 cm (B) 17.5 cm (C) 20.7 cm (D) 72.0 cm

Open solution

Q11. NEET 2024

A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?



(A) Point P moves slower than point Q

(B) Point P moves faster than point Q

(C) Both the points P and Q move with equal speed

(D) Point P has zero speed

Open solution

Q12. NEET 2023 Manipur

A constant torque of $100\ \mathrm{N\ m}$ turns a wheel of moment of inertia $300\ \mathrm{kg\ m^2}$ about an axis passing through its centre. Starting from rest, its angular velocity after $3\ \mathrm{s}$ is :-

(A) 1 rad/s (B) 5 rad/s (C) 10 rad/s (D) 15 rad/s

Open solution

Q13. NEET 2023

The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is :-

(A) $5:3$ (B) $2:5$ (C) $\sqrt{5}:\sqrt{3}$ (D) $\sqrt{3}:\sqrt{5}$

Open solution

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Q14. NEET 2023

The angular acceleration of a body, moving along the circumference of a circle, is :

(A) along the radius towards the centre
(B) along the tangent to its position
(C) along the axis of rotation
(D) along the radius, away from centre

Open solution

Q15. NEET 2022 Phase 2

An energy of 484 J is spent in increasing the speed of a flywheel from 60 rpm to 360 rpm. The moment of inertia of the flywheel is :

(A) 0.07 kg-m^2
(B) 0.7 kg-m^2
(C) 3.22 kg-m^2
(D) 30.8 kg-m^2

Open solution

Q16. NEET 2022 Phase 1

The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s^2 is

(A) 2π
(B) 4π
(C) 12π
(D) 104π

Open solution

Q17. NEET 2022 Phase 1

The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is

(A) 2 : 1
(B) $\sqrt{2}$: 1
(C) 4 : 1
(D) 1 : $\sqrt{2}$

Open solution

Q18. NEET 2021

From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the center of the ring and perpendicular to the plane of the ring is 'K' times 'MR^2'. Then the value of 'K' is :

(A) $\frac{1}{8}$
(B) $\frac{3}{4}$
(C) $\frac{7}{8}$
(D) $\frac{1}{4}$

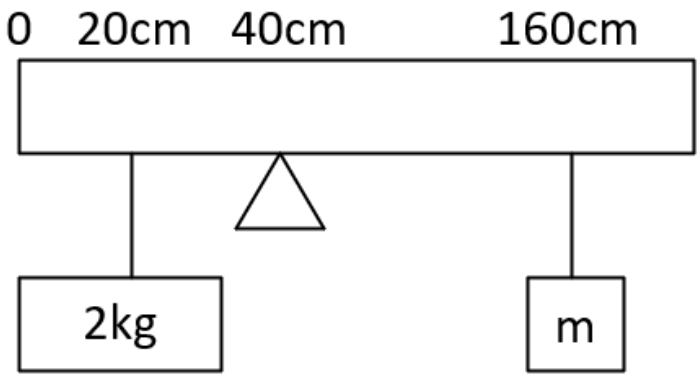
Open solution

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Q19. NEET 2021

A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. ($g = 10\ m/s^2$)

Open solution



(A) $\frac{1}{12}$ kg (B) $\frac{1}{2}$ kg (C) $\frac{1}{3}$ kg (D) $\frac{1}{6}$ kg

Q20. NEET 2020 Phase 1

Find the torque about the origin when a force of 3 $\hat{j}$ N acts on a particle whose position vector is 2 $\hat{k}$ m.

(A) 6 $\hat{j}$ N m (B) $-$ 6 $\hat{i}$ N m (C) 6 $\hat{k}$ N m (D) 6 $\hat{i}$ N m

Open solution

Q21. NEET 2019

A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is :

(A) 2×10^{-3} N m (B) 2×10^{-6} N m (C) 2×10^{6} N m (D) 12×10^{-4} N m

Open solution

Q22. NEET 2019

Two particles A and B are moving in uniform circular motion in concentric circles of radii r_A and r_B with speed v_A and v_B respecitively. Their time period of rotation is the same. The ratio of angular speed of A to that of B will be :

(A) $v_A : v_B$ (B) 1 : 1 (C) $r_B : r_A$ (D) $r_A : r_B$

Open solution

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Q23. NEET 2019

A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it?

(A) 1 J (B) 2 J (C) 3 J (D) 30 kJ

Open solution

Q24. NEET 2018

A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is

(A) 7 : 10 (B) 5 : 7 (C) 10 : 7 (D) 2 : 5

Open solution

Q25. NEET 2018

A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere?

(A) Angular velocity
(B) Moment of inertia
(C) Rotational kinetic energy
(D) Angular momentum

Open solution

Q26. NEET 2018

Three objects, A : (a solid sphere), B : (a thin circular disk) and C : (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation

(A) $W_C > W_B > W_A$ (B) $W_A > W_B > W_C$ (C) $W_B > W_A > W_C$ (D) $W_A > W_C > W_B$

Open solution

Q27. NEET 2018

The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at (2, 0, −3), about the point (2, −2, −2), is given by

(A) $-8\hat{i} - 4\hat{j} - 7\hat{k}$
(B) $-4\hat{i} - \hat{j} - 8\hat{k}$
(C) $-7\hat{i} - 8\hat{j} - 4\hat{k}$
(D) $-7\hat{i} - 4\hat{j} - 8\hat{k}$

Open solution

Q28. NEET 2017

A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

(A) 0.25 rad s^{-2} (B) 25 rad s^{-2} (C) 5 m s^{-2} (D) 25 m s^{-2}

Open solution

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Q29. NEET 2017

Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2. They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is

Open solution

(A) $\frac{1}{4}I(\omega_1 - \omega_2)^2$
(B) $I(\omega_1 - \omega_2)^2$
(C) $\frac{1}{8}I(\omega_1 - \omega_2)^2$
(D) $\frac{1}{2}I(\omega_1 + \omega_2)^2$

Q30. NEET 2016 Phase 2

Two rotating bodies A and B of masses m and 2m with moments of inertia I_A and I_B ($I_B > I_A$) have equal kinetic energy of rotation. If L_A and L_B be their angular momenta respectively, then

Open solution

(A) $L_A = \frac{L_B}{2}$
(B) $L_A = 2L_B$
(C) $L_B > L_A$
(D) $L_A > L_B$

Q31. NEET 2016 Phase 2

A solid sphere of mass m and radius R is rotating about its diameter. A solid cylinder of same mass and same radius is also rotating about its geometrical axis with an angular speed twice that of the sphere. The ratio of their kinetic energies of rotation (E_{sphere} / $E_{cylinder}$) will be

Open solution

(A) 2 : 3
(B) 1 : 5
(C) 1 : 4
(D) 3 : 1

Q32. NEET 2016 Phase 2

A light rod of length l has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is

Open solution

(A) $\frac{m_1 m_2}{m_1 + m_2} l^2$
(B) $\frac{m_1 + m_2}{m_1 m_2} l^2$
(C) $(m_1 + m_2) l^2$
(D) $\sqrt{m_1 m_2}\, l^2$

Q33. NEET 2016 Phase 1

A disc and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?

Open solution

(A) Both reach at the same time
(B) Depends on their masses
(C) Disc
(D) Sphere

Q34. NEET 2016 Phase 1

From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre?

Open solution

(A) 11 $MR^2/32$
(B) 9 $MR^2/32$
(C) 15 $MR^2/32$
(D) 13 $MR^2/32$

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Q35. NEET 2016 Phase 1

A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad s^{-2}. Its net acceleration in m s^{-2} at the end of 2.0 s is approximately

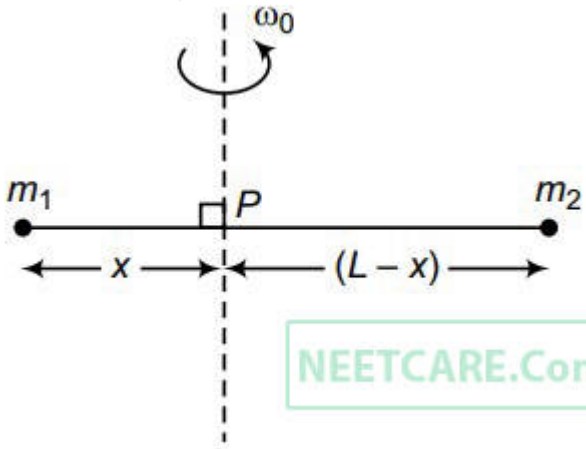
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(A) 6.0 (B) 3.0 (C) 8.0 (D) 7.0

Q36. AIPMT 2015

Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L, and negligible mass. The rod is to be set rotating about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotating with angular velocity ω_0 is minimum, is given by

Open solution



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(A) $x = \frac{m_2}{m_1}L$ (B) $x = \frac{m_2 L}{m_1+m_2}$
(C) $x = \frac{m_1 L}{m_1+m_2}$ (D) $x = \frac{m_1}{m_2}L$

Q37. AIPMT 2015

An automobile moves on a road with a speed of 54 km h^{-1}. The radius of its wheels is 0.45 m and the moment of inertia of the wheel about its axis of rotation is 3 kg m^2. If the vehicle is brought to rest in 15 s, the magnitude of average torque transmitted by its brakes to the wheel is

Open solution

(A) 10.86 kg m^2 s^{-2} (B) 2.86 kg m^2 s^{-2} (C) 6.66 kg m^2 s^{-2} (D) 8.58 kg m^2 s^{-2}

Q38. AIPMT 2015

A force $\vec{F} = \alpha\hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of α for which angular momentum about origin is conserved is

Open solution

(A) zero (B) 1 (C) -1 (D) 2

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Q39. AIPMT 2015 Cancelled Paper

A rod of weight W is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at distance x from A. The normal reaction on A is

Open solution

(A) $\frac{W(d-x)}{x}$

(B) $\frac{W(d-x)}{d}$

(C) $\frac{Wx}{d}$

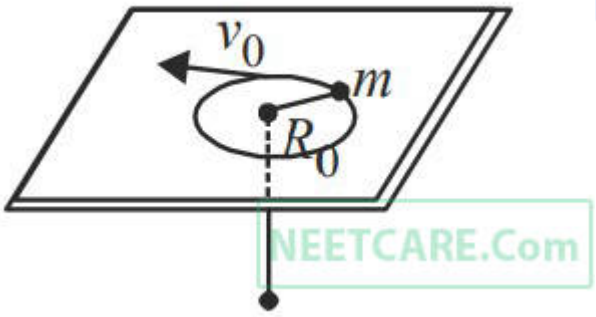
(D) $\frac{Wd}{x}$

Q40. AIPMT 2015 Cancelled Paper

A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0. The mass is attached to a string which passes through a smooth hole in the plane as shown. the tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$.

Open solution

The final value of the kinetic energy is



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(A) $2mv_0^2$

(B) $\frac{1}{2}mv_0^2$

(C) mv_0^2

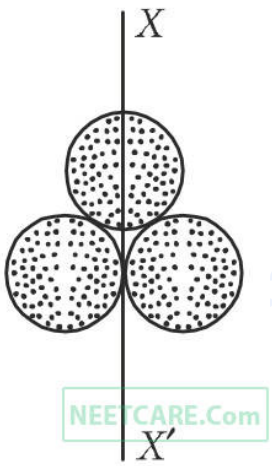
(D) $\frac{1}{4}mv_0^2$

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Q41. AIPMT 2015 Cancelled Paper

Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' which is touching to two shells and passing through diameter of third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is

Open solution



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(A) $\frac{16}{5}mr^2$ (B) $4mr^2$ (C) $\frac{11}{5}mr^2$ (D) $3mr^2$

Q42. AIPMT 2014

A solid cylinder of mass 50 kg and radius 0.5 m is free to rotate about the horizontal axis. A massless string is wound round the cylinder with one end attached to it and other hanging freely. Tension in the string required to produce an angular acceleration of 2 revolutions s^{-2} is

Open solution

(A) 25 N (B) 50 N (C) 78.5 N (D) 157 N

Q43. AIPMT 2014

The ratio of the accelerations for a solid sphere (mass m and radius R) rolling down an incline of angle θ without slipping and slipping down the incline without rolling is

Open solution

(A) 5 : 7 (B) 2 : 3 (C) 2 : 5 (D) 7 : 5

Q44. NEET 2013 (Karnataka)

The ratio of radii of gyration of a circular ring and a circular disc, of the same mass and radius, about an axis passing through their centres and perpendicular to their planes are

Open solution

(A) $1 : \sqrt{2}$ (B) 3 : 2 (C) 2 : 1 (D) $\sqrt{2} : 1$

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Q45. NEET 2013 (Karnataka)

Two discs are rotating about their axes, normal to the discs and passing through the centres of the discs. Disc D_1 has 2 kg mass and 0.2 m radius and initial angular velocity of 50 rad s^{-1}. Disc D_2 has 4 kg mass, 0.1 m radius and initial angular velocity of 200 rad s^{-1}. The two discs are brought in contact face to face, with their axes of rotation coincident. The final angular velocity (in rad s^{-1}) of the system is

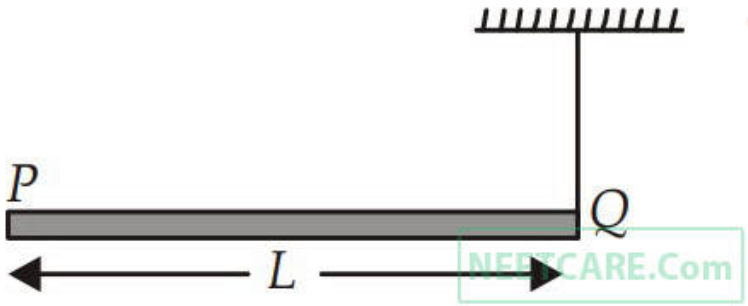
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(A) 60 (B) 100 (C) 120 (D) 40

Q46. NEET 2013

A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is

Open solution



(A) $\frac{2g}{L}$ (B) $\frac{2g}{2L}$ (C) $\frac{3g}{2L}$ (D) $\frac{g}{L}$

Q47. NEET 2013

A small object of uniform density rolls up a curved surface with an initial velocity 'v'. It reaches upto a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is

Open solution

(A) hollow sphere (B) disc (C) ring (D) solid sphere

Q48. AIPMT 2012 Mains

A car of mass m is moving on a level circular track of radius R. If μ_s represents the static friction between the road and tyres of the car, the maximum speed of the car in circular motion is given by

Open solution

(A) $\sqrt{\mu_s mRg}$ (B) $\sqrt{\frac{Rg}{\mu_s}}$

(C) $\sqrt{\frac{mRg}{\mu_s}}$ (D) $\sqrt{\mu_s Rg}$

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Q49. AIPMT 2012 Mains

A circular platform is mounted on a frictionless vertical axle. Its radius R = 2 m and its moment of inertia about the axle is 200 kg m^2. It is initially at rest. A 50 kg man stands on the edge of the platform and begins to walk along the edge at the speed of 1 ms^{-1} relative to the ground. Time taken by the man to complete one revolution is

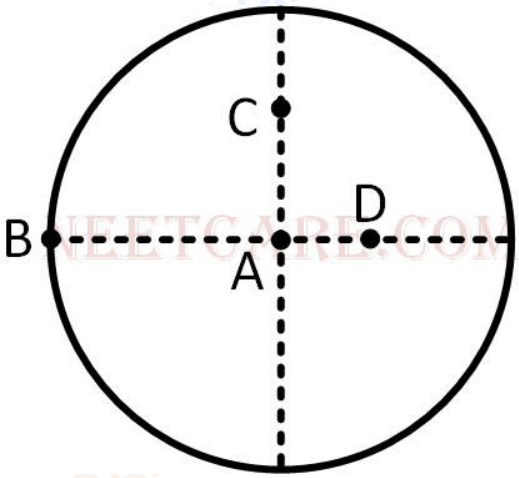
Open solution

(A) π s (B) $\frac{3\pi}{2}$ s (C) 2π s (D) $\frac{\pi}{2}$ s

Q50. AIPMT 2012 Mains

The moment of inertia of a uniform circular disc is maximum about an axis perpendicular to the disc and passing through

Open solution



(A) B (B) C (C) D (D) A

Q51. AIPMT 2012 Prelims

When a mass is rotating in a plane about a fixed point, its angular momentum is directed along

Open solution

(A) a line perpendicular to the plane of rotation

(B) the line making an angle of 45° to the plane of rotation

(C) the radius

(D) the tangent to the orbit

Q52. AIPMT 2012 Prelims

A car of mass 1000 kg negotiates a banked curve of radius 90 m on a frictionless road. If the banking angle is 45°, the speed of the car is

Open solution

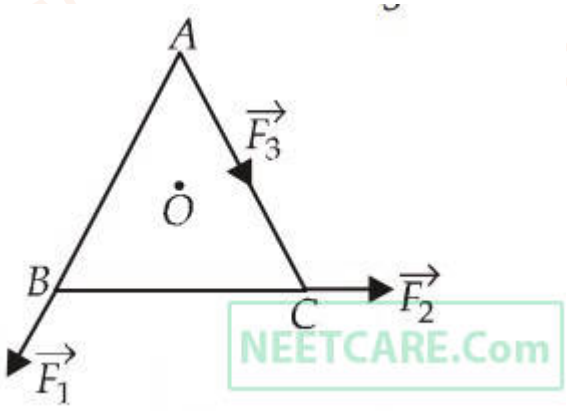
(A) 20 m s^{-1} (B) 30 m s^{-1} (C) 5 m s^{-1} (D) 10 m s^{-1}

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Q53. AIPMT 2012 Prelims

Open solution

ABC is an equilateral triangle with O as its centre. $\vec{F_1}, \vec{F_2}$ and $\vec{F_3}$ represent three forces acting along the sides AB, BC and AC respectively. If the total torque about O is zero then magnitude of $\vec{F_3}$ is

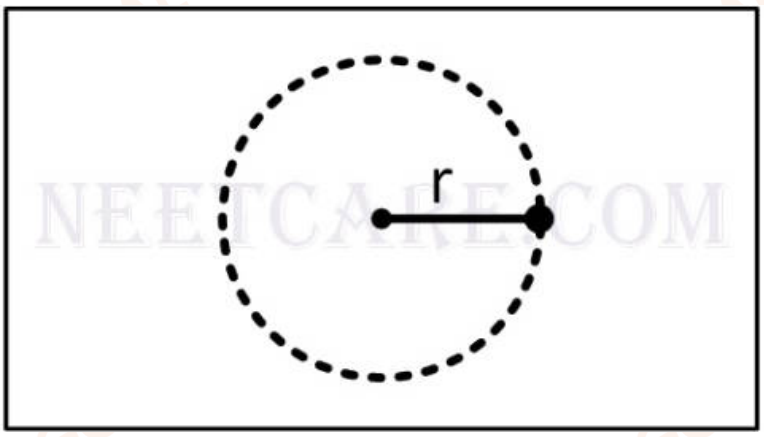


(A) $F_1 + F_2$ (B) $F_1 - F_2$ (C) $\frac{F_1+F_2}{2}$ (D) $2(F_1 + F_2)$

Q54. AIPMT 2011 Mains

Open solution

A small mass attached to a string rotates on a frictionless table top as shown. If the tension in the string is increased by pulling the string causing the radius of the circular motion to decrease by a factor of 2, the kinetic energy of the mass will



(A) decrease by a factor of 2
(B) remain constant
(C) increase by a factor of 2
(D) increase by a factor of 4

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Q55. AIPMT 2011 Prelims

The instantaneous angular position of a point on a rotating wheel is given by the equation $\theta(t) = 2t^3 - 6t^2$

The torque on the wheel becomes zero at

(A) t = 1 s (B) t = 0.5 s (C) t = 0.25 s (D) t = 2 s

Open solution

Q56. AIPMT 2011 Prelims

The moment of inertia of a thin uniform rod of mass M and length L about an axis passing through its midpoint and perpendicular to its length is I_0. Its moment of inertia about an axis passing through one of its ends and perpendicular to its length is

(A) $I_0 + ML^2/2$ (B) $I_0 + ML^2/4$ (C) $I_0 + 2ML^2$ (D) $I_0 + ML^2$

Open solution

Q57. AIPMT 2010 Mains

From a circular disc of radius R and mass 9M, a small disc of mass M and radius $\frac{R}{3}$ is removed concentrically. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through its centre is

(A) $\frac{40}{9}$ MR^2 (B) MR^2 (C) 4MR^2 (D) $\frac{4}{9}$ MR^2

Open solution

Q58. AIPMT 2010 Mains

A solid cylinder and a hollow cylinder, both of the same mass and same external diameter are released from the same height at the same time on an inclined plane. Both roll down without slipping. Which one will reach the bottom first?

(A) Both together only when angle of inclination of plane is 45°

(B) Both together

(C) Hollow cylinder

(D) Solid cylinder

Open solution

Q59. AIPMT 2010 Mains

A thin circular ring of mass M and radius r is rotating about its axis with constant angular velocity ω. Two objects each of mass m are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with angular velocity given by

(A) $\frac{(M+2m)\omega}{2m}$ (B) $\frac{2M\omega}{M+2m}$

(C) $\frac{(M+2m)\omega}{M}$ (D) $\frac{M\omega}{M+2m}$

Open solution

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Q60. AIPMT 2010 Prelims

A circular disk of moment of inertia I_t is rotating in a horizontal plane, about its symmetry axis, with a constant angular speed ω_i. Another disk of moment of inertia I_b is dropped coaxially onto the rotating disk. Initially the second disk has zero angular speed. Eventually both the disks rotate with a constant angular speed ω. The energy lost by the initially rotating disc to friction is

(A) $\frac{1}{2}\frac{I_b^2}{(I_t+I_b)}\omega_i^2$
(B) $\frac{1}{2}\frac{I_t^2}{(I_t+I_b)}\omega_i^2$
(C) $\frac{I_b-I_t}{(I_t+I_b)}\omega_i^2$
(D) $\frac{1}{2}\frac{I_bI_t}{(I_t+I_b)}\omega_i^2$

Open solution

Q61. AIPMT 2010 Prelims

A gramophone record is revolving with an angular velocity ω. A coin is placed at a distance r from the centre of the record. The static coefficient of friction is μ. The coin will revolve with the record if

(A) r = $\mu g\omega^2$
(B) r < $\frac{\omega^2}{\mu g}$
(C) $r \leq \frac{\mu g}{\omega^2}$
(D) $r \geq \frac{\mu g}{\omega^2}$

Open solution

Q62. AIPMT 2009

A thin circular ring of mass M and radius R is rotating in a horizontal plane about an axis vertical to its plane with a constant angular velocity ω. If two objects each of mass m be attached gently to the opposite ends of a diameter of the ring, the ring will then rotate with an angular velocity

(A) $\frac{\omega M}{M+2m}$
(B) $\frac{\omega(M+2m)}{M}$
(C) $\frac{\omega M}{M+m}$
(D) $\frac{\omega(M-2m)}{M+2m}$

Open solution

Q63. AIPMT 2009

If $\vec{F}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then

(A) $\vec{r}.\vec{\tau} > 0$ and $\vec{F}.\vec{\tau} < 0$
(B) $\vec{r}.\vec{\tau} = 0$ and $\vec{F}.\vec{\tau} = 0$
(C) $\vec{r}.\vec{\tau} = 0$ and $\vec{F}.\vec{\tau} \neq 0$
(D) $\vec{r}.\vec{\tau} \neq 0$ and $\vec{F}.\vec{\tau} = 0$

Open solution

Q64. AIPMT 2009

Four identical thin rods each of mass M and length l, form a square frame. Moment of inertia of this frame about an axis through the centre of the square and perpendicular to its plane is

(A) $\frac{2}{3}Ml^2$
(B) $\frac{13}{3}Ml^2$
(C) $\frac{1}{3}Ml^2$
(D) $\frac{4}{3}Ml^2$

Open solution

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Q65. AIPMT 2008

A thin rod of length L and mass M is bent at its midpoint into two halves so that the angle between them is 90°. The moment of inertia of the bent rod about an axis passing through the bending point and perpendicular to the plane defined by the two halves of the rod is

Open solution

(A) $\frac{ML^2}{6}$ (B) $\frac{\sqrt{2}ML^2}{24}$ (C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{12}$

Q66. AIPMT 2008

The ratio of the radii of gyration of a circular disc to that of a circular ring, each of same mass and radius, around their respective axes is

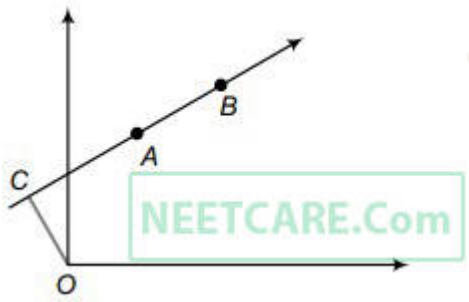
Open solution

(A) $\sqrt{2}:1$ (B) $\sqrt{2}:\sqrt{3}$ (C) $\sqrt{3}:\sqrt{2}$ (D) $1:\sqrt{2}$

Q67. AIPMT 2007

A particle of mass m moves in the XY plane with a velocity v along the straight line AB. If the angular momentum of the particle with respect to origin O is L_A when it is at A and L_B when it is at B, then

Open solution



(A) $L_A = L_B$

(B) the relationship between L_A and L_B depends upon the slope of the line AB

(C) $L_A < L_B$

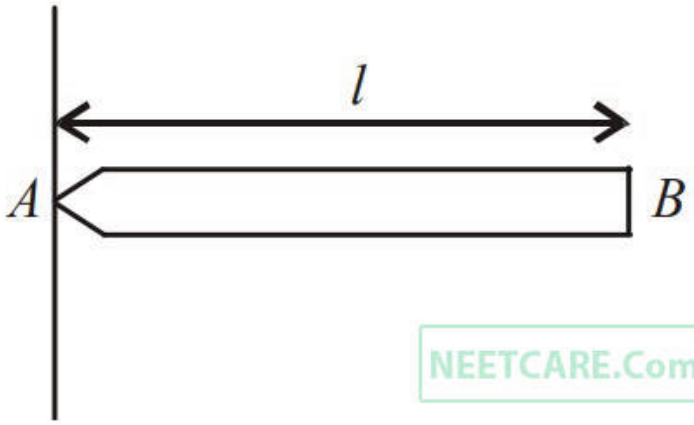
(D) $L_A > L_B$.

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Q68. AIPMT 2007

A uniform rod AB of length l and mass m is free to rotate about point A. The rod is released from rest in the horizontal position. Given that the moment of inertia of the rod about A is $ml^2/3$, the initial angular acceleration of the rod will be

Open solution



NEETCARE.Com

(A) $\frac{mgl}{2}$ (B) $\frac{3}{2}gl$ (C) $\frac{3g}{2l}$ (D) $\frac{2g}{3l}$

Q69. AIPMT 2007

A wheel has angular acceleration of 3.0 rad/sec^2 and an initial angular speed of 2.00 rad/sec. In a time of 2 sec it has rotated through an angle (in radian) of

Open solution

(A) 10 (B) 12 (C) 4 (D) 6

Q70. AIPMT 2006

A uniform rod AB of length l and mass m is free to rotate about point A. The rod is released from rest in the horizontal position. Given that the moment of inertia of the rod about A is $ml^2/3$, the initial angular acceleration of the rod will be

Open solution

(A) $\frac{mgl}{2}$ (B) $\frac{3}{2}gl$ (C) $\frac{3g}{2l}$ (D) $\frac{2g}{3l}$

Q71. AIPMT 2006

The moment of inertia of a uniform circular disc of radius R and mass M about an axis touching the disc at its diameter and normal to the disc

Open solution

(A) $\frac{1}{2}MR^2$ (B) MR^2 (C) $\frac{2}{5}MR^2$ (D) $\frac{3}{2}MR^2$

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Q72. AIPMT 2006

A tube of length L is filled completely with an incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω. The force exerted by the liquid at the other end is

(A) $\frac{ML^2\omega^2}{2}$ (B) $\frac{ML\omega^2}{2}$ (C) $\frac{ML^2\omega}{2}$ (D) $ML\omega^2$

Open solution

Q73. AIPMT 2005

The moment of inertia of a uniform circular disc of radius R and mass M about an axis passing from the edge of the disc and normal to the disc is

(A) MR^2 (B) $\frac{1}{2}MR^2$ (C) $\frac{3}{2}MR^2$ (D) $\frac{7}{2}MR^2$

Open solution

Q74. AIPMT 2005

A drum of radius R and mass M, rolls down without slipping along an inclined plane of angle θ. The frictional force

(A) dissipates energy as heat

(B) decreases the rotational motion

(C) decreases the rotational and translation motion

(D) converts translational energy to rotational energy.

Open solution

Q75. AIPMT 2005

Two bodies have their moments of inertia I and 2I respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular velocity will be in the ratio

(A) 2 : 1 (B) 1 : 2 (C) $\sqrt{2}:1$ (D) $1:\sqrt{2}$

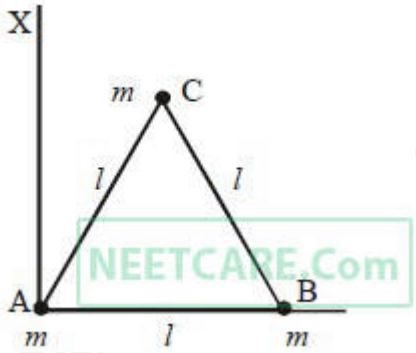
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Q76. AIPMT 2004

Three particles, each of mass m gram, are situated at the vertices of an equilateral triangle ABC of side l cm (as shown in the figure). The moment of inertia of the system about a line AX perpendicular to AB and in the plane of ABC, in gram-cm^2 units will be

Open solution



(A) $\frac{3}{4}ml^2$ (B) $2ml^2$ (C) $\frac{5}{4}ml^2$ (D) $\frac{3}{2}ml^2$

Q77. AIPMT 2004

A wheel having moment of inertia 2 kg m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in one minute would be

Open solution

(A) $\frac{2\pi}{15}$ N m (B) $\frac{\pi}{12}$ N m (C) $\frac{\pi}{15}$ N m (D) $\frac{\pi}{18}$ N m

Q78. AIPMT 2004

A round disc of moment of inertia I_2 about its axis perpendicular to its plane and passing through its centre is placed over another disc of moment of inertia I_1 rotating with an angular velocity ω about the same axis. The final angular velocity of the combination of discs is

Open solution

(A) $\frac{I_2\omega}{I_1+I_2}$ (B) ω
(C) $\frac{I_1\omega}{I_1+I_2}$ (D) $\frac{(I_1+I_2)\omega}{I_1}$

Q79. AIPMT 2004

The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axes in the plane of the ring is

Open solution

(A) 2 : 3 (B) 2 : 1 (C) $\sqrt{5} : \sqrt{6}$ (D) $1 : \sqrt{2}$

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Q80. AIPMT 2003

A stone is tied to a string of length l and is whirled in a vertical circle with the other end of the string as the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of the change in velocity as it reaches a position where the string is horizontal (g being acceleration due to gravity) is

(A) $\sqrt{2\,(u^2 - gl)}$ (B) $\sqrt{u^2 - gl}$

(C) $u - \sqrt{u^2 - 2gl}$ (D) $\sqrt{2gl}$

Open solution

Q81. AIPMT 2003

A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K. If radius of the ball be R. then the fraction of total energy associated with its rotational energy will be

(A) $\frac{K^2+R^2}{R^2}$ (B) $\frac{K^2}{R^2}$ (C) $\frac{K^2}{K^2+R^2}$ (D) $\frac{R^2}{K^2+R^2}$

Open solution

Q82. AIPMT 2003

A solid cylinder of mass M and radius R rolls without slipping down an inclined plane of length L and height h. What is the speed of its centre of mass when the cylinder reaches its bottom ?

(A) $\sqrt{2gh}$ (B) $\sqrt{\frac{3}{4}gh}$ (C) $\sqrt{\frac{4}{3}gh}$ (D) $\sqrt{4gh}$

Open solution

Q83. AIPMT 2003

A thin circular ring of mass M and radius r is rotating about its axis with a constant angular velocity ω. Four objects each of mass m, are kept gently to the opposite ends of two perpendicular diameters of the ring. The angular velocity of the ring will be :

(A) $\frac{M\omega}{4m}$ (B) $\frac{M\omega}{M+4m}$

(C) $\frac{(M+4m)\omega}{M}$ (D) $\frac{(M-4m)\omega}{M+4m}$

Open solution

Q84. AIPMT 2002

A point P consider at contact point of a wheel on ground which rolls on ground without slipping then value of displacement of point P when wheel completes half of rotation (if radius of wheel is 1 m)

(A) 2 m (B) $\sqrt{\pi^2 + 4}m$ (C) $\pi\, m$ (D) $\sqrt{\pi^2 + 2}\, m$

Open solution

Q85. AIPMT 2002

A solid sphere of radius R is placed on smooth horizontal surface. A horizontal force F is applied at height h from the lowest point. For the maximum acceleration of centre of mass, which is correct ?

(A) h = R (B) h = 2R

(C) h = 0 (D) no relation between h and R.

Open solution

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Q86. AIPMT 2002

A disc is rotating with angular speed ω. If a child sits on it, what is conserved

(A) linear momentum
(B) angular momentum
(C) kinetic energy
(D) potential energy.

Open solution

Q87. AIPMT 2002

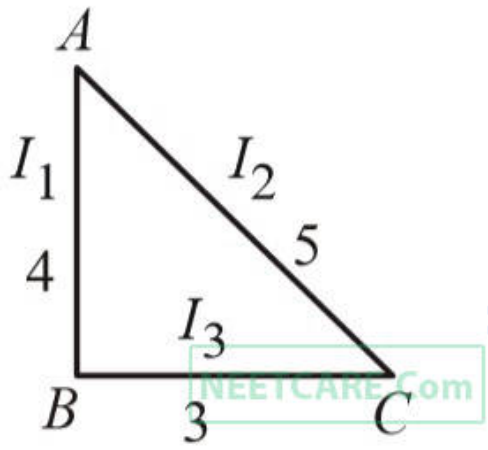
A circular disc is to be made by using iron and aluminium so that it acquired maximum moment of inertia about geometrical axis. It is possible with

(A) aluminium at interior and iron surround to it
(B) iron at interior and aluminium surround to it
(C) using iron and aluminium layers in alternate order
(D) sheet of iron is used at both external surface and aluminium sheet as internal layers.

Open solution

Q88. AIPMT 2000

For the adjoining diagram, the correct relation between I_1, I_2, and I_3 is, (I-moment of inertia)



(A) $I_1 > I_2$
(B) $I_2 > I_1$
(C) $I_3 > I_1$
(D) $I_3 > I_2$

Open solution

Q89. AIPMT 2000

For a hollow cylinder and a solid cylinder rolling without slipping on an inclined plane, then which of these reaches earlier

(A) solid cylinder
(B) hollow cylinder
(C) both simultaneously
(D) can't say anything.

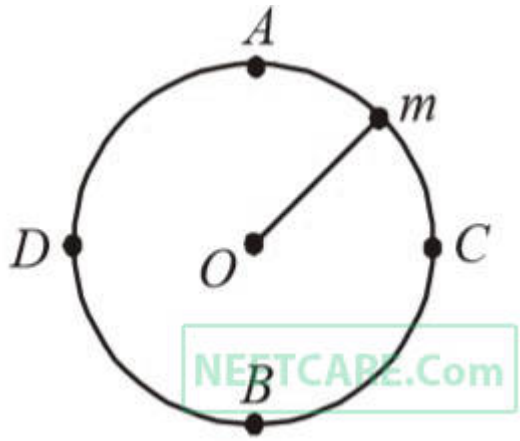
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Q90. AIPMT 2000

As shown in the figure at point O a mass is performing vertical circular motion. The average velocity of the particle is increased, then at which point will the string break

Open solution



(A) A (B) B (C) C (D) D

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Answer Key

1. **D**	2. **C**	3. **C**	4. **B**	5. **D**
6. **B**	7. **A**	8. **D**	9. **B**	10. **A**
11. **B**	12. **A**	13. **D**	14. **C**	15. **B**
16. **B**	17. **B**	18. **B**	19. **A**	20. **B**
21. **B**	22. **B**	23. **C**	24. **B**	25. **D**
26. **A**	27. **D**	28. **B**	29. **A**	30. **C**
31. **B**	32. **A**	33. **D**	34. **D**	35. **C**
36. **B**	37. **C**	38. **C**	39. **B**	40. **A**
41. **B**	42. **D**	43. **A**	44. **D**	45. **B**
46. **C**	47. **B**	48. **D**	49. **C**	50. **A**
51. **A**	52. **B**	53. **A**	54. **D**	55. **A**
56. **B**	57. **A**	58. **D**	59. **D**	60. **D**
61. **C**	62. **A**	63. **B**	64. **D**	65. **D**
66. **D**	67. **A**	68. **C**	69. **A**	70. **C**
71. **D**	72. **B**	73. **C**	74. **D**	75. **C**
76. **C**	77. **C**	78. **C**	79. **C**	80. **A**
81. **C**	82. **C**	83. **B**	84. **B**	85. **D**
86. **B**	87. **A**	88. **B**	89. **A**	90. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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